NEURAL SPACETIME SEARCH WITH GRTECLYN: A PRACTICAL BASELINE FOR DYNAMICAL METRIC DISCOVERY

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We outline a practical baseline for turning `GRTeclyn` into a closed-loop GPU search engine for exotic spacetime metrics. A Python scheduler proposes compact initial-data parameters, a C++/AMReX executable evolves each candidate, and scoring diagnostics promote only geometries that survive, show operational faster-than-light (FTL) benefit, and avoid coordinate artifacts. Matter is handled either by algebraic reconstruction from the Einstein constraints at $t = 0$ or, in the long-term path, by an elliptic positive-energy initial-data solve.

I. INTRODUCTION

Most proposed interstellar metrics are hand-designed static ansätze. That approach is analytically clean but strongly biased toward highly symmetric geometries.

Here we treat metric discovery as black-box optimization over dynamical 3D initial data, searching for novel faster-than-light spacetime geometries—encompassing FTL transport, FTL communication, portal-like shortcuts, and wormhole throats—rather than restricting attention to any single warp ansatz. The framework couples a Python optimizer to the GPU AMR evolution engine in `GRTeclyn`, with explicit checks for matter consistency, stability, and fake-FTL gauge effects. A successful candidate must survive dynamical evolution, show an operational shortcut relative to a flat-spacetime control, keep tidal forces bounded, and minimize violations of the standard energy conditions.

Blind random 3D initial data almost always violates constraints, collapses, or relaxes to flat space. Optimization supplies the steering: maximize FTL benefit while penalizing constraint error, negative energy, tidal force, and instability.

This fits a broader shift toward automated theory exploration [1–4]. For warp geometries, WARP FACTORY and WARPAX emphasize observer-robust energy-condition evaluation [5, 6]; we extend that philosophy from analysis of prescribed metrics to search over evolved initial data, while keeping numerical convergence and constraint satisfaction central [7, 8].

Throughout this paper, "FTL" means effective faster-than-light propagation through a curved spacetime geometry compared with flat-spacetime light travel. It does not imply local superluminal motion relative to the local light cone.

II. INITIAL DATA

The optimizer proposes the spatial metric γ_{ij} and extrinsic curvature K_{ij}; the Hamiltonian and momentum constraints then determine the compatible energy density ρ and momentum density j_i:

$$\rho = \frac{R + K^2 - K_{ij}K^{ij}}{16\pi}, \qquad j_i = \frac{1}{8\pi} D_j \left(K_i^j - \delta_i^j K \right). \tag{1}$$

Regions with $\rho < 0$ are recorded as exotic-matter cost and fed back into the objective. A future positive-energy path using an elliptic constraint solver is discussed in Sec. VII.

III. CANDIDATE PARAMETERIZATION

The optimizer emits a compact vector θ; a decoder maps it to smooth AMReX initial fields.

A. Level 1: Radial Recipes

The initial fields at $t = 0$ are built from spherically symmetric radial profiles:

$$q(r) = q_\infty + \sum_n a_n B_n(r), \tag{2}$$

where q denotes χ, α, β^i, or K, and B_n are smooth radial basis functions.

B. Level 2: Angular Recipes (implemented)

Angular modes add non-spherical structure:

$$q(r, \theta, \varphi) = q_\infty + \sum_{n\ell m} a_{n\ell m} B_n(r) Y_{\ell m}(\theta, \varphi). \tag{3}$$

This is implemented for gauge fields (α, β^i), where angular modes carry no constraint cost; geometric angular modes require the future 3D constraint solver.

C. Search space

The three FTL shortcut channels (shift-driven warp, $\chi > 1$ compression, areal-radius throat pinching) are independent. The default search vector is nine-dimensional:

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four χ coefficients, a shared basis width, two lapse coefficients, and two shift coefficients—all dimensionless amplitudes for the Gaussian basis expansion. This spans all three mechanisms while remaining tractable for CMA-ES. Higher-level parameterizations (neural decoders, LLM meta-agents) are deferred to Sec. VII.

IV. THE SEARCH LOOP

The pipeline keeps Python orchestration separate from precompiled GPU evolution. The Python controller writes `params.txt` and dispatches Slurm tasks; GPU nodes run the same `GRTeclyn` binary for every candidate, avoiding recompilation inside the search loop.

Each optimizer evaluation follows:

1. propose basis coefficients θ;
2. derive constrained fields, write `params.txt`, and reject bad pre-flight residuals;
3. evolve on GPU and write constraint, collapse, matter, and plotfile diagnostics;
4. parse diagnostics into $S(\theta)$ and update CMA-ES after each generation.

The optimizer sees only the scalar black-box objective $f(\theta) = -S(\theta)$, making the score design the central physics interface.

Tier 1 uses short low-resolution single-GPU runs with early stopping; Tier 2 promotes survivors to multi-GPU resolution ladders for stability and convergence.

The current pipeline uses CMA-ES, not reinforcement learning: it samples continuous coefficients, ranks them by $-S(\theta)$, and updates a Gaussian search distribution. CMA-ES cannot add mouths, throats, or new mode families; RL becomes relevant only for sequential decisions such as choosing topology, modes, or mid-run gauge actions. A future hierarchical search could let a discrete controller choose topology and active modes, then hand the resulting skeleton to CMA-ES/MAP-Elites for continuous refinement, echoing discrete/continuous decompositions in string-landscape search [3].

V. SCORE COMPONENTS AND WHY THEY MATTER

The central challenge is to discover spacetime metrics that exhibit specific physical qualities: effective FTL shortcuts, dynamical stability, bounded tidal forces, and minimal exotic matter. The score function must reward these qualities while ensuring that trivial Minkowski spacetime—which is perfectly stable and constraint-clean—does not dominate the search. This section defines every component of the scoring objective in one place.

A. Bounded reward and total score

Each raw diagnostic value v_k (such as a constraint norm, a growth rate, or an exotic-matter integral) is mapped to a bounded reward $s_k \in [0, 1]$ via

$$s_k = r(v_k; \sigma_k) = \frac{1}{1 + v_k/\sigma_k}, \tag{4}$$

where $v_k \geq 0$ is the diagnostic value (lower is better for penalties) and σ_k is a characteristic scale that controls sensitivity—when $v_k = \sigma_k$, the reward is exactly $\frac{1}{2}$. The total score is the weighted sum

$$S(\theta) = \sum_k w_k\, s_k(\theta). \tag{5}$$

This mapping keeps every component in $[0, 1]$ regardless of physical units, while the weights w_k (Table I) set the relative importance.

B. FTL shortcut metrics

Three independent shortcut indicators are evaluated on the x-axis from the $t = 0$ profiles.

a. Warp null-ray shortcut. For conformally flat slices, an axial null ray has coordinate speed $dx/dt = \alpha(x)\sqrt{\chi(x)} - \beta^x(x)$, giving travel time

$$T_{\text{curved}} = \int_{-L}^{L} \frac{dx}{\alpha\sqrt{\chi} - \beta^x}, \qquad T_{\text{flat}} = 2L, \tag{6}$$

and reward

$$F_{\text{null}} = \max\left(0,\, \frac{T_{\text{flat}} - T_{\text{curved}}}{T_{\text{flat}}}\right). \tag{7}$$

Paths below a coordinate-speed floor are rejected.

b. Portal proper-distance shortcut. The portal-like compression reward is

$$D_{\text{proper}} = \int_{-L}^{L} \chi^{-1/2}(r)\, dr,$$
$$F_{\text{portal}} = \max\left(0,\, \frac{2L - D_{\text{proper}}}{2L}\right). \tag{8}$$

c. Wormhole throat pinch. The throat reward compares an interior areal-radius minimum to the asymptotic radius:

$$F_{\text{throat}} = \max\left(0,\, 1 - \frac{R_{\text{min}}}{R_{\text{asymp}}}\right). \tag{9}$$

d. Anti-flatness gate and combined FTL objective. The anti-flat gate ensures only non-trivial geometries score:

$$s_{\text{nonflat}} = \min\left(1,\, \frac{\max|\chi - 1| + \max|\beta^x|}{\tau}\right), \tag{10}$$

with scale $\tau \sim 0.08$, giving

$$F_{\text{FTL}} = \max(F_{\text{null}}, F_{\text{portal}}, F_{\text{throat}}) \cdot s_{\text{nonflat}}. \tag{11}$$

e. Natário–Alcubierre expansion asymmetry. For a warp-like contraction-ahead/expansion-behind pattern:

$$F_{\text{asymmetry}} = \max\left(0,\ \frac{\int_{x>0} -\theta(\mathbf{x})\, d^3x + \int_{x<0} \theta(\mathbf{x})\, d^3x}{\int |\theta(\mathbf{x})|\, d^3x + \epsilon_\theta}\right). \tag{12}$$

The implemented 1D evaluator reduces these integrals to trapezoidal sums along the travel axis.

f. Logarithmic FTL amplification. Weak shortcuts are log-amplified so even a few-percent shortcut becomes optimizer-visible:

$$F_{\log} = \frac{\ln(1 + \eta F_{\text{FTL}})}{\ln(1 + \eta)}, \tag{13}$$

with $\eta \approx 100$. The optimizer uses $F_{\log}$ rather than raw F_{FTL}.

g. Operational probe FTL. A major risk is gauge-driven fake FTL. High-scoring candidates must therefore pass an operational arrival-time test against a flat control on the same grid, with no trapped-surface crossing and bounded constraints:

$$F_{\text{op}} = \frac{t_B^{\text{flat}} - t_B^{\text{curved}}}{t_B^{\text{flat}}}, \tag{14}$$

accepted only when constraints remain bounded and no trapped surface is crossed.

C. Health and stability metrics

a. Co-moving stability. Eulerian stability penalizes moving bubbles, so stability is estimated in a co-moving frame. The average axial shift is

$$\langle \beta^x \rangle = \frac{1}{V_{\text{bubble}}} \int_{\text{bubble}} \beta^x(\mathbf{x})\, d^3x, \tag{15}$$

and the co-moving budget is

$$\begin{aligned} \Delta_{\text{comoving}} &= \frac{\max_t |\partial \chi / \partial t|}{\max(|\langle \beta^x \rangle|,\ 10^{-3})}, \\ s_{\text{comoving}} &= \frac{1}{1 + \Delta_{\text{comoving}}}. \end{aligned} \tag{16}$$

For stationary throats or portals with negligible shift, the scorer falls back to Eulerian stability.

b. Dynamical stability and geometry growth. Short runs can survive while already focusing toward collapse. We therefore score geometry growth explicitly. For each diagnostic q (from $\alpha_{\min}$, $\chi_{\min}$, $|K|_{\max}$, r_H, R_{areal}), fractional violation terms are

$$\begin{aligned} \Delta q_{\text{grow}} &\equiv \frac{\max_t q(t) - q(0)}{\max\big(|q(0)|,\ q_{\text{floor}}\big)}, \\ \Delta q_{\text{drop}} &\equiv \frac{q(0) - \min_t q(t)}{\max\big(|q(0)|,\ q_{\text{floor}}\big)}, \end{aligned} \tag{17}$$

with terms clipped to ≥ 0. The implemented indicators are:

$$\Delta_K \equiv \Delta\big(|K|_{\max}\big)_{\text{grow}} \quad \text{with } q_{\text{floor}} = 10^{-1}, \tag{18}$$

$$\Delta_\alpha \equiv \Delta\big(\alpha_{\min}\big)_{\text{drop}} \quad \text{with } q_{\text{floor}} = 10^{-6}, \tag{19}$$

$$\Delta_\chi \equiv \Delta\big(\chi_{\min}\big)_{\text{drop}} \quad \text{with } q_{\text{floor}} = 10^{-8}, \tag{20}$$

$$\Delta_{r_H} \equiv \Delta\big(r_H\big)_{\text{grow}} \quad \text{with } q_{\text{floor}} = 1, \tag{21}$$

$$\Delta_{R_{\text{areal}}} \equiv \max_t \frac{\big|R_{\text{areal}}(t) - R_{\text{areal}}(0)\big|}{\max\big(|R_{\text{areal}}(0)|,\ 10^{-8}\big)}. \tag{22}$$

The geometry-change budget $\Delta_{\text{geom}} = \Delta_K + \Delta_\alpha + \Delta_\chi + \Delta_{r_H} + \Delta_{R_{\text{areal}}}$ gives the stability reward

$$s_{\text{stability}} = r(\Delta_{\text{geom}};\, 1) = \frac{1}{1 + \Delta_{\text{geom}}}. \tag{23}$$

Higher is better; $s_{\text{stability}} < 0.25$ is treated as a promotion failure. A good equilibrium should approach $s_{\text{stability}} \gtrsim 0.8$.

c. Constraint growth rate. Late-time diagnostics are fit to $q(t) \sim q_0 e^{\lambda t}$ and the worst growth is penalized:

$$\begin{aligned} s_{\text{growth}} &= r(\max(0, \lambda_{\text{eff}});\, \sigma_\lambda)\,, \\ \lambda_{\text{eff}} &= \max\big(\lambda_{\|H\|},\ \lambda_{|K|_{\max}},\ \lambda_{1/\chi_{\min}}\big), \end{aligned} \tag{24}$$

with $\sigma_\lambda = 0.5$.

D. Risk penalties

a. ANEC line proxy. The required energy is integrated along the travel axis:

$$\begin{aligned} \mathcal{A}_{\text{line}} &= \int_{-L}^{L} \rho_{\text{req}}(x)\, dx, \\ s_{\text{anec}} &= r(\max(0, -\mathcal{A}_{\text{line}});\, \sigma_{\mathcal{A}})\,, \end{aligned} \tag{25}$$

with $\rho_{\text{req}} = (R + \frac{2}{3}K^2)/16\pi$ and $\sigma_{\mathcal{A}} = 0.1$. The full averaged null energy condition target is

$$\mathcal{A} = \int_\gamma T_{\mu\nu}\, k^\mu k^\nu\, d\lambda, \tag{26}$$

which requires the evolved probe and a local tetrad.

b. Tidal proxy. $T_{\text{tidal}} = \max_x |R(x)| + \max_x |\partial_x^2 \alpha(x)|$, combined with a $t = 0$ trapped-surface flag until full 4-curvature diagnostics are available.

c. Other risk terms. The score also includes s_{horizon} (penalizing trapped-surface formation), s_{EC} (penalizing energy-condition violations from the evolved stress-energy), and $s_{\text{exotic}}^{\text{eff}}$ (a Hawking–Ellis/Type-I hybrid margin from the evolved 4-metric following WARPAX [6], computed with fourth-order finite differences [7]).

E. The non-triviality gate

Flat Minkowski spacetime scores perfectly on every health metric. To prevent it from dominating, health rewards are multiplied by a non-triviality gate:

$$g_{\rm nt} = \min\Big(1,\ \max\big(s_{\rm nonflat},\, F_{\rm asymmetry},\, s_{\rm nontrivial},\, F_{\rm curv},\, F_{\rm op},\, F_{\rm log},\, s^{\rm eff}_{\rm exotic}\big)\Big) \in [0,1]. \tag{27}$$

The health rewards $\mathcal{H}$ are gated, while FTL/exoticity terms remain ungated:

$$S(\theta) = g_{\rm nt}(\theta)\sum_{k\in\mathcal{H}} w_k s_k(\theta) \ + \sum_{k\notin\mathcal{H}} w_k s_k(\theta). \tag{28}$$

With this gate, flat Minkowski drops to $S = 0$ while structured reference geometries keep their original scores. Unavailable diagnostics contribute zero, so legacy episodes re-score consistently.

F. Constraint-satisfying recipe

For the conformally flat recipe ($\tilde{\gamma}_{ij} = \delta_{ij}$, $A_{ij} = 0$), the constraints reduce to

$$R[\chi] + \tfrac{2}{3}K^2 = 16\pi\rho, \tag{29}$$

$$-\tfrac{2}{3}D^i K = 8\pi j^i. \tag{30}$$

With $\Pi = 0$, the momentum constraint requires spatially constant K; the scalar field ϕ is then derived from χ to reduce Hamiltonian violation. The pipeline consists of `constrained_recipe.py` (derive ϕ from χ with K = const, $\Pi = 0$), `preflight.py` (reject large residuals or negative χ before GPU launch), and `RadialRecipeLevel.cpp` (write $\rho_{\rm req}$ and constraint diagnostics).

A test suite passes **74/74 assertions**: flat space remains exact, $K = \text{const}, \Pi = 0$ drives $\|M_i\|_{L^2} = 0$, pathological inputs are rejected, and constrained overrides usually reduce $\|H\|_{L^2}$. On 27 synthetic profiles, the batch harness finds 7 `accepted_phantom`, 19 `rejected_exotic_energy`, and 1 `rejected_negative_chi`.

The first symmetry-breaking extension adds axisymmetric deformations,

$$\chi(r,\theta) = \chi_0(r) + \sum_a A_a\, e^{-(r-r_a)^2/2w_a^2}\, P_{\ell_a}(\cos\theta), \tag{31}$$

with a ray check to ensure $\chi > 0$.

VI. RESULTS

A. Reference seed validation

The FTL scoring formulas were validated on analytical reference seeds at $L = 8$. Flat Minkowski scores $F_{\rm FTL} = 0$ across all channels. The logarithmic amplification makes subluminal Alcubierre signals optimizer-visible ($F_{\rm log} = 0.43$–0.57 for $v = 0.3$–0.8), while Ellis–Bronnikov and Schwarzschild score through the throat and proper-distance channels ($F_{\rm FTL} > 0.87$) even with $F_{\rm null} = 0$. On the risk side, Alcubierre seeds have zero ANEC violation and zero tidal curvature at $t = 0$ (pure-shift warp), whereas Ellis–Bronnikov and Schwarzschild carry large axis exotic-energy integrals ($\mathcal{A}_{\rm line} = -6.6$ and -2.4) and trapped-surface flags.

B. GPU smoke validation

Short $t = 2$ GPU runs at $N_{\rm full} = 64$ validated the end-to-end pipeline. Three spherical candidates survived with controlled 3D constraint norms ($\|H\|_{L^2} \le 0.042$), confirming that the 1D pre-flight is conservative. Stability scores ranged from $s_{\rm stability} = 0.30$ (Ellis–Bronnikov) to 0.68 (bubble-wall profile). Extended $t = 5$ tests promoted Ellis–Bronnikov and the bubble-wall seed to the resolution ladder.

C. Non-spherical candidates

All seven ray-sane non-spherical candidates (dipole, quadrupole, mixed-mode, and random angular deformations) survived short GPU evolution with controlled constraints ($\|H\|_{L^2} \le 0.014$, $s_{\rm stability} \approx 0.44$–$0.51$). However, these χ-only angular deformations produce strong expansion asymmetry ($F_{\rm asym} \approx 0.99$) but zero axial null shortcut ($F_{\rm log} = 0$). Angular χ deformations plateau on asymmetry alone; shift-enabled warps are needed to exceed flat space.

D. Stability and growth analysis

At $t = 2$, the upgraded FTL scoring resolves the flat-space attractor: Alcubierre $v = 0.5$ ($S = 10.08$) narrowly outranks flat Minkowski ($S = 10.00$) because $F_{\rm log}$ and $s_{\rm comoving}$ offset the Eulerian stability penalty, while Ellis–Bronnikov scores highest ($S = 10.84$) through its throat channel.

However, every short-run survivor still has positive exponential growth rate $\lambda_{\rm eff}$, ranging from 0.29 (random seeds) to 1.02 (Ellis–Bronnikov), exposing the central stability gap: short-run survival does not imply long-term equilibrium. The first full-domain AMR CMA-ES campaign confirmed this—all top candidates had $s_{\rm stability} < 0.09$, showing that the optimizer exploits the short stop time rather than finding genuinely stable geometries. This motivates the $s_{\rm growth}$ penalty and longer-run validation below.

TABLE I. Compact score-weight summary. Components are inactive when their diagnostics are unavailable.

Group	Weights
FTL value	$5F_{\mathrm{log}} + 2F_{\mathrm{asymmetry}} + s_{\mathrm{nonflat}} + 3F_{\mathrm{op}}$
Health	$1.5s_{\mathrm{survival}} + 2s_{\mathrm{constraints}} + s_{\mathrm{lapse}} + 2.5s_{\mathrm{comoving}} + 0.5s_{\mathrm{stability}}$
Risk penalties	$1.5s_{\mathrm{horizon}} + 2s_{\mathrm{EC}} + 2s_{\mathrm{growth}} + 1.5s_{\mathrm{anec}} + s_{\mathrm{tidal}}$
Evolved geometry	$0.5F_{\mathrm{curv}} + 1.5s^{\mathrm{eff}}_{\mathrm{exotic}}$

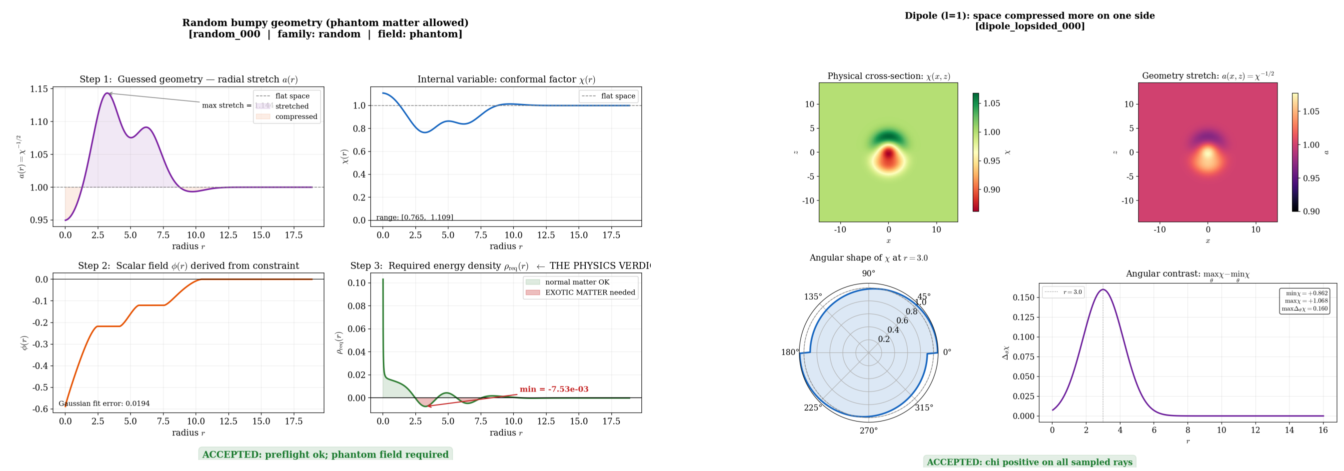

FIG. 1. **Metric-guesser validation examples.** Left: accepted phantom random candidate, showing guessed stretch $a(r) = \chi^{-1/2}$, stored $\chi(r)$, derived scalar $\phi(r)$, and required energy density $\rho_{\mathrm{req}}(r)$; it is numerically constructible but accepted only in the phantom/exotic-matter class because $\rho_{\mathrm{req}} < 0$ in places. Right: accepted non-spherical dipole candidate; the $\ell = 1$ mode makes the geometry lopsided while χ stays positive throughout the ray check.

E. Longer-run validation

Longer $t = 16$ runs (Table II) test the exotic scalar matter fix and evolved operational FTL probe beyond the smoke window. Three main implications emerge. First, the exotic field produces the expected matter NEC violations for Ellis–Bronnikov and mild non-spherical phantom candidates. Second, effective energy-condition evaluation of $T^{\mathrm{eff}}_{\mu\nu} = G_{\mu\nu}/8\pi$ recovers the geometric exotic energy of shift-driven Alcubierre data even when scalar-matter NEC is zero. Third, evolved-frame FTL is essential: Alcubierre's initial shortcut decays, while Ellis–Bronnikov develops an evolved shortcut absent on the $t = 0$ slice. The gated objective removes flat Minkowski ($S = 0$) without changing structured candidates.

F. Search optimization

Beyond CMA-ES, two complementary strategies were tested. A surrogate-assisted RBF kernel-ridge model prescreens proposals, skipping 27% of GPU evaluations in one 8-GPU run while retaining the best score. MAP-Elites over FTL benefit and exoticity produces a diverse failure atlas (7 distinct elites from 40 evaluations on a 6×6 grid, scores $S = 9.9$–12.5) rather than a single optimum. Pareto extraction over $(F_{\mathrm{FTL}}, s_{\mathrm{anec}}, s_{\mathrm{growth}}, s_{\mathrm{tidal}})$ exposes trade-offs hidden by scalar weights.

G. The dipole metric: gauge-angular search winner

Because angular χ deformations cannot be made constraint-consistent by a 1D solve, we opened the gauge sector: α and β^i carry Legendre modes with $\ell \in \{1, 2\}$ and searchable radial centers/widths, giving a 21-parameter gauge-angular space. An 8-GPU campaign found evolved superluminal channels at $\int \rho_- \, dV \approx 0.13$–$0.17$, a factor 2–5 below spherical persistent-FTL candidates.

The top candidate, `eval_000188`, was promoted and re-evolved to $t = 16$ at $N_{\mathrm{full}} = 192$, `max_level=2` (Table III, Fig. 2). The validated geometry is a standing Alcubierre-like χ dipole, not a translating bubble: the evolved channel persists weakly by $t = 16$, constraints decrease, and exotic matter remains small and axial, but a near-trapped flag appears in the compressed lobe.

The dipole metric achieves a gated total score of $S = 16.75$, outperforming both Alcubierre $v = 0.5$ ($S = 14.9$) and Ellis–Bronnikov ($S = 12.8$) at $t = 16$. Crucially, it accomplishes this with an exotic-matter cost of $\int \rho_- \, dV = 0.134$, substantially lower than spherical persistent-FTL candidates, while maintaining a superluminal path fraction of 0.51 in the evolved frame. This

TABLE II. Longer-run validation ($t = 16$, $N_{\rm full} = 64$, $L = 8$, H100) with the gated objective (Eq. 28). $F^0_{\rm op}/F^{\rm ev}_{\rm op}$ are the operational FTL benefits on the $t = 0$ slice and the evolved plotfile; $\mathrm{NEC}^{\rm mat}$ is the minimum matter null-energy margin over the run (negative = violation); $g_{\rm nt}$ is the non-triviality gate. S is the gated total: flat Minkowski collapses to 0 while every non-trivial geometry is unchanged. All reference runs use constrained phantom initial data.

Example	survival	$s_{\rm stab}$	$F^0_{\rm op}$	$F^{\rm ev}_{\rm op}$	$\mathrm{NEC}^{\rm mat}$	$g_{\rm nt}$	S
Flat Minkowski	1.00	1.00	0	—	0	0.0	**0.0**
Alcubierre $v = 0.5$	1.00	0.02	0.096	0.0005	0	1.0	**14.9**
Ellis–Bronnikov (N=16)	0.80	0.08	0	0.042	−0.62	1.0	12.8
`quadrupole_bubble_001`	1.00	0.18	0	0.0095	−0.013	1.0	10.5
Schwarzschild puncture	0.29	0.00	0	—	−0.91	1.0	9.7

TABLE III. High-quality streaming validation of the gauge-angular winner `eval_000188` ($N_{\rm full} = 192$, `max_level=2`, $t = 16$, H100). Initial vs. evolved operational FTL, exotic matter, and constraint health.

Quantity	Value
$\max c$ (initial slice)	2.67
$\max c$ (evolved, $t = 16$)	1.32
$F^0_{\rm op}$ / $F^{\rm ev}_{\rm op}$	0.35 / 5.6×10^{-3}
superluminal path fraction (evolved)	0.51
$\int \rho_- \, dV$ (exotic matter)	0.134
ANEC line integral (axis)	-6.6×10^{-3}
$\Vert H\Vert_{L^2}$ (max / final)	1.1×10^{-3} / 4.2×10^{-4}
$\Vert M\Vert_{L^2}$ (final)	3.6×10^{-5}
$\min \theta_+$ (near-trapped flag)	−0.11
S (gated total)	16.75

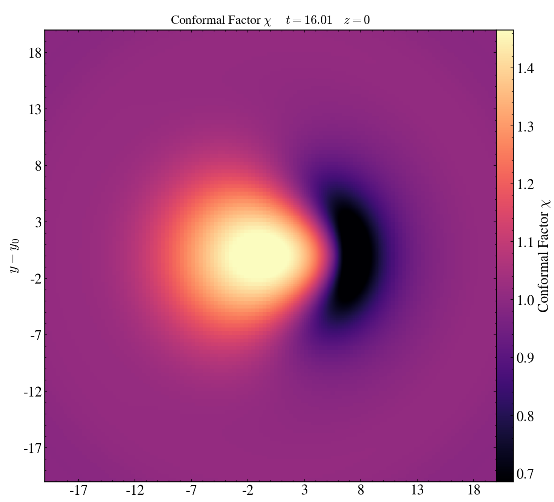

FIG. 2. **Persistent directional channel of the gauge-angular winner** `eval_000188` (conformal factor χ, $z = 0$ slice, $t = 16$, $N_{\rm full} = 192$). A standing front–back dipole compresses space on the $+x$ side ($\chi \approx 0.7$, dark) and expands it behind ($\chi \approx 1.45$, bright)—an Alcubierre-like contract-ahead/expand-behind polarity realized through the shift sector at low exotic-matter cost. The structure does not translate; it stands and slowly decays.

demonstrates that the automated search can discover geometries that outperform hand-designed warp metrics on the combined objective of FTL benefit, stability, and energy-condition cost.

VII. DISCUSSION AND FUTURE WORK

This work turns exotic-metric study into a closed-loop GPU search over dynamical initial data. The current pipeline proposes constrained recipes, evolves them in `GRTeclyn`, scores stability/FTL/energy diagnostics, and validates promising candidates on longer runs. The first campaigns expose the central stability gap—optimizers can still favor geometries that merely outlive a short stop time—but also show that FTL scaffolding, non-triviality gating, growth metrics, evolved energy-condition checks, surrogate screening, and MAP-Elites can steer the search away from trivial flat space toward genuinely structured geometries like the dipole metric.

Several extensions would substantially expand the search capabilities.

a. Elliptic constraint solver. The largest single improvement is making the constraint-satisfying path the default. A positive-energy elliptic solver (e.g. `GRTresna`) would deform each proposed geometry until the Hamiltonian and momentum constraints are satisfied with $\rho \geq 0$ before evolution, reducing junk radiation and making growth rates cleaner. This would also allow directional structure in χ or $\tilde{A}_{ij}$ without reintroducing off-axis constraint error.

b. Topology search. The current parameterization cannot change topology: its non-spherical freedom lives in the gauge sector, where shift structure naturally favors warp-like channels. Wormholes and portals require a discrete controller that selects mouths, throats, modes, and boundary connectivity before the continuous optimizer refines each topology, echoing discrete/continuous decompositions in string-landscape search [3].

c. Additional degrees of freedom. Gauge-angular search produced a low-exotic directional channel. Next degrees of freedom include searchable radial placement of angular modes, full-z domains without reflective parity, persistence/co-motion rewards, soft barriers on near-trapped surfaces, and higher angular order with a dedicated translation shift.

d. Advanced parameterization. Future Level 3 replaces basis functions with an implicit neural decoder, optionally trained with Hamiltonian and momentum residuals as PINN losses [8]. A rule-based spacetime grammar can enforce asymptotic flatness, positivity, $K = \text{const}$, $\Pi = 0$, and chosen topology before evaluation, reducing wasted GPU jobs in the spirit of grammar-constrained theory generation [1]. Level 4 is an LLM meta-agent that classifies failed runs, retunes exposed gauge/grid parameters, and proposes new modes while leaving the physics verdict to read-only diagnostics [2].

e. Multi-observer energy conditions. The $t = 0$ ANEC and tidal proxies are blind to pure-shift warps. Evaluating energy conditions from the evolved 4-metric via $T_{\mu\nu} = G_{\mu\nu}/8\pi$, e.g.

$$\Xi_W(\mathbf{x}) = \min_V T_{\mu\nu} V^\mu V^\nu, \qquad g_{\mu\nu} V^\mu V^\nu = -1, \tag{32}$$

recovers Alcubierre NEC/WEC violations missed by χ-sourced proxies. A Hawking–Ellis/Type-I hybrid margin following WARPAX [6] with fourth-order finite differences [7] feeds the score as $s^{\text{eff}}_{\text{exotic}}$.

f. Production-scale search. With surrogate screening, MAP-Elites, and Pareto extraction in place, production runs target a 1000-candidate constraint atlas plus a 50-generation CMA-ES campaign.

g. Analytical back-derivation. A numerical optimum is only a coefficient list. The final handoff reconstructs χ_{num}, fits a closed form by symbolic regression, solves the Hamiltonian constraint for $\phi(r)$, and reloads the analytic profile for verification. This can be automated as a multi-agent loop inspired by PhysAgent [4].

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